University of New Hampshire
Peter T. Paul College of Business and Economics

ECON 898/720: MACHINE LEARNING FOR ECONOMICS

TERM 3, SPRING, 2026

Course Information:

Course ID:	ECON 898/720	**Credit Hours:**	3.00 for 898 /4.00 for 720
Department:		Economics	
Attributes:		Graduate level course	
Start/End Dates:		1/20/26 to 3/13/26	
Class Times/Location:		MW 5:10 pm - 7:00 pm @ Paul205	
Recommended Readings: (Optional)		1.Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow *by* Aurelien Geron 2. Causal Inference and Discovery in Python *by* Aleksander Molak, Ajit Jaokar	

Instructor Information:

Instructor:	Bingjin (Ben) Xue Assistant Professor of Economics
Office Location/Phone:	Paul College 321 (603) 862- 3364
Office Hours:	MW 1:00 pm-2:00pm or by appointment
Email Address **Preferred Method of Contact:**	bingjin.xue@unh.edu Discussion forum on MyCourses are preferred contact for course-related questions and I will respond within one business day.

MACHINE LEARNING FOR ECONOMICS

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Course Overview

Machine Learning for Economics bridges economic reasoning with the computational power of AI. Rather than treating machine learning as a black box, this course opens it—training students to think like economists who engineer algorithms with purpose and interpretability. Students will learn to use Python to build, test, and explain machine learning models grounded in economic theory and data. The goal is to equip future economists with the skills to not only apply machine learning responsibly, but also to extend and adapt these methods in pursuit of rigorous economic research and policy evaluation.

Prerequisites

This course utilizes math skills including calculus and optimization.
Undergraduate students should have a grade of B- or above in the following (or similar) courses: ECON 606 - Intermediate Microeconomics with Calculus; ECON 726 - Introduction to Econometrics.
Graduate students should have a grade of B- or above in the following (or similar) courses: ECON 926 - Econometrics I

Learning Objectives

By course end it is expected that you will be able to:

- Get familiar with Python and Jupyter Notebook
- Use Python to discover and visualize the data and gain economic insights.
- Handle different types of economic data and build data processing pipelines.
- Build machine learning models to make predictions and causal inferences.
- Fine-tune machine learning models.

Textbook & Software:

Recommended Optional Textbooks
1.Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow by Aurelien Geron
2. Causal Inference and Discovery in Python *by* Aleksander Molak, Ajit Jaokar

Note: The course primarily follows a set of lecture notes created by the professor. While textbooks are not required, they can serve as useful references for certain topics. The professor will highlight the relevant chapters in the textbooks when introducing specific subjects.

Software
No required software, we will use Google Colab

Note: The course utilizes Google Colab, an online platform for running Python scripts. If you prefer to work locally, I recommend using Anaconda with Jupyter Lab as your IDE.

Learning Management Systems:

MyCourses is the learning management tool we use for this course. The site will be used to post announcements, maintain a gradebook and provide a discussion forum.

Grades & Grading Scale

The course grade consists of the following components:

3 Homework Assignments	20% * 3 = 60%
1 In-class Final Project	30%
Participation (in the form of in-class quizzes)	10%
Total	**100%**

** Participation is assessed by random in-class quizzes

The following scale, which is the standard UNH grading scale, determines letter grades. Grades do NOT get rounded up.

Percentage	Grade	Percentage	Grade
94–100	A	77–79.99	C+
90–93.99	A-	74-76.99	C
87–89.99	B+	70–73.99	C-
84-86.99	B	69.99 or lower	D or F
80–83.99	B-		

Assignment Details

- **Assignments:** Every other week there will be one coding assignment on MyCourses, which is due one week after. Assignments may be submitted up to 7 days late, but for each day past the deadline that you take, you will incur a 10% penalty.[1]
  Graduate students will be asked to answer one more theoretical question in the assignments.

- **Final Project (in-class):** There is one comprehensive final project, in which you will use all the techniques you've learned in the class to analyze a dataset and draw insights.
  The final project has different expectations for undergraduate and graduate students. Graduate students should expect to have extra questions and carry out more rigorous statistical inference in the final project.

[1] For example, if you would have earned an 88% but you submitted 2 days late, the grade would be a 68%.

- **Participation:** You will frequently be asked to respond to progress assessments during lecture days. These assessments account for 10% of your total grade and cannot be made up. The participation final grade is averaged over all assessments given in the semester.

- **Missed Class:** If you must miss class for any reason, it is your responsibility to ensure you keep up with material and turn in assignments on time. I strongly recommend that you review the relevant material on the MyCourses site have another student provide you with a scan or copy of their notes from the session you missed. Please contact your instructor with any questions about the material or assignments you missed.

Course Workload and Credit Hours

This syllabus reflects the federal definition of a credit hour, which entails a minimum 3 hours of engaged time per week per credit over a 15-week semester (student workload for shorter length terms must be increased proportionally per week to maintain required approximately 45 hours of work per credit per term). Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours. Student work reflects intended learning outcomes and is verified through evidence of student achievement.

The undergraduate version of the course (ECON 720) is 4 credits to reflect additional nongraded assignments and guided learning aimed at reinforcing fundamental concepts. These assignments will be given *ad hoc* throughout the semester.

Course Schedule

Week	Topic	Assignment
	First Day of Term 3	
1	Python, Jupyter Notebook, Colab	nongraded reading
	Descriptive Analysis: Tables and Graphs - matplotlib	
	Descriptive Analysis: Statistics - Numpy, pandas	
2	Descriptive Analysis: Distribution - scipy, seaborn	**Assignment 1**
	Python – if, for, functions, modules	
	Regression Model- statsmodels	
3	Model Specification: Bias-Variance Instrumental Variable Models	nongraded reading
	Hypothesis Test	

4	Machine Learning: a project - scikit-learn	**Assignment 2**
	Supervised Learning: Linear Models	
	Training Models: Computational Complexity, Gradient Descent, Learning Curves	
5	Supervised Learning: Decision Trees/Causal Trees	nongraded reading
	Ensembles: Gradient boosting, random forest, bagging, voting, stacking	
	Supervised Learning: Other popular models: SVM, NB, LDA	
6	Unsupervised Learning: K-Means, NN	**Assignment 3**
	Unsupervised Learning: Gaussian Mixture	
7	Model Selection and evaluation: Cross-validation, Metrics and scoring	nongraded reading
	Tuning the hyper-parameters of an estimator	
8	Neural Networks and Deep Learning -PyTorch	
	Computer Vision Using CNN	
	Causal Machine Learning – an introduction	
	Last Day of Term 3	**Final Project***

***** The final project has different expectations for undergraduate and graduate students. Graduate students should expect to have extra questions and carry out more rigorous statistical inference in the final project.

Student to Instructor Communication Expectations

How to Reach Me

Discussion Forum:

A Discussions tab can be found on MyCourses. Questions related to assignments or learning should first be asked here, if possible, so that the answers are visible to all other students. I will be active in the MyCourses class area daily, Monday through Friday. If you post a question for me in the discussion forum, please anticipate a response within 24 hours or sooner.

Email:

Bingjin.Xue@unh.edu

The use of email is reserved for questions of a private nature. If you would like to send me an email, please anticipate a response within 48 hours.

In-person Office hours:

MW 1:00-2:00 pm at Paul 321

Technical Requirements and Technical Support

See website listings for current recommendations and requirements related to this course - https://online.unh.edu/technical-requirements. Technical assistance related to Canvas is available at https://itsupport.unh.edu/mycourses/.

Academic Honesty and Plagiarism

Students are required to abide by the UNH Academic Honesty policy located in the Student Rights, Rules, and Responsibilities Handbook.

As your instructor, I proactively monitor academic integrity through regular use of tools like Turnitin and a diversified assessment approach. All works submitted to SafeAssign become a part of a UNH proprietary database. This is actively used to identify future intellectual property theft. Plagiarism of any type may be grounds for receiving an "F" in an assignment or an "F" in the overall course. Plagiarism is defined as "the unattributed use of the ideas, evidence, or words of another person, or the conveying the false impression that the arguments and writing in a paper are your own." (UNH Academic Honesty Policy, 09.3) Examples of plagiarism include:

• Copying part or all of a previously published or written document, without quotation marks. This cut-and-paste practice is considered plagiarism, even if the writing is cited properly.
• Copying the sentence and/or paragraph structure of a source while substituting other words. "Patchwriting" is very common among students who struggle to understand information. Whether or not it is cited, this kind of paraphrasing is considered plagiarism, and it indicates that students have not fully understood the information.
• Submitting work that was completed for other courses, other times, by other students or the student him/herself.
• Submitting writing acquired from a "paper mill "resource.
UNH provides a comprehensive Plagiarism Tutorial for students who may be unclear about what specifically constitutes "plagiarism".

Incidents of academic dishonesty are reported to the school dean and may be grounds for further action. You can contact me at any time on this issue. Additional resources are located below:

http://libraryguides.unh.edu/unhmcitingsources
http://www.library.unh.edu/reference/citation.shtml
UNH Academic Honesty Policy, 09.3

Confidentiality and Mandatory Reporting of Sexual Violence or Harassment

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's Title IX

Coordinator (Bo Zaryckyj, Bo.Zaryckyj@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence and harassment shared by students. If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy at (603) 862-7233/TTY (800) 735-2964. For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting options), please visit student reporting options and resources easily accessible on their phones.
Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu, or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY / 7-1-1 Relay NH, or visit the CREO website.
Anonymous reports may be submitted.

Emotional or mental health distress

Your academic success and overall mental health are very important. If, during the semester, you find you are experiencing emotional or mental health issues, please contact the University's (PACS) (3rd floor, Smith Hall; 603-862-2090/TTY: 7-1-1; https://www.unh.edu/pacs/) which provides counseling appointments and other mental health services. If urgent, students may call PACS M-F, 8 a.m.-5 p.m., and schedule an Urgent Same-Day Appointment.

University Disability Accommodations

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS); 227 Smith Hall, or sas.office@unh.edu.
Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS, but are under no obligation to provide retroactive accommodations.
For more information, contact SAS: 227 Smith Hall, www.unh.edu/sas, 603.862.2607, 711 (Relay NH) or sas.office@unh.edu

Note: **This syllabus is subject to change. Students will be promptly notified of any changes.**